Neuromorphic Computing for Intelligent Battery Management Systems: A Comprehensive Review of Spiking Neural Networks, Reservoir Computing and Edge AI

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***Abstract*—Lithium-ion battery packs sit at the centre of electric vehicles and grid-scale energy storage. Every safety, control and prognostic function of a Battery Management System (BMS) depends on accurate, real-time estimates of State of Charge (SOC), State of Health (SOH), State of Power (SOP) and Remaining Useful Life (RUL), none of which can be measured directly. This review traces the field from classical model-based estimators (Coulomb counting, OCV mapping, ECMs, Kalman-filter variants) through conventional deep learning (RNNs, LSTMs, Transformers) to neuromorphic computing, Spiking Neural Networks (SNNs), reservoir computing, and Liquid State Machines (LSMs), together with the hardware (Intel Loihi 2, SpiNNaker2, BrainChip Akida, IBM TrueNorth, and FPGA platforms) and event-driven sensing schemes required to deploy them at the edge. A multi-dimensional benchmarking framework demonstrates that while neuromorphic paradigms achieve order-of-magnitude energy efficiency on sparse battery signals, standardized hardware benchmarking is the next critical milestone.

***Keywords*—Battery Management System; State of Charge; State of Health; Spiking Neural Networks; Reservoir Computing; Liquid State Machine; Neuromorphic Hardware; Edge AI.**

I. EXECUTIVE SUMMARY & BENCHMARKING TAXONOMY

Modern BMS hardware is bounded by strict thermal envelopes, microampere quiescent currents, and low-cost microcontrollers. Traditional deep neural networks provide high predictive accuracy but their dense matrix-multiplication demands outstrip automotive MCU budgets. Neuromorphic computing offers event-driven, temporally native computation well suited for quasi-static battery sensor telemetry.

Estimator Paradigm	Typical Model	Inference Energy	MCU SRAM Footprint	Dynamic Adaptability	Edge Feasibility
Classical Observer	2-RC EKF / UKF	~50 uJ	< 4 KB	Moderate (needs RLS)	High (Production standard)
Deep Learning	LSTM / GRU / Transformer	~5 mJ	> 128 KB	Low (needs retrain)	Low (requires NPU/DSP)
Spiking Neural Net	LIF / Spike-Reservoir	~5 uJ	~16 KB	Moderate (STDP)	High on Neuromorphic ASIC
Reservoir Computing	Leaky-Integrator ESN	~12 uJ	< 12 KB (with CSR)	High (Online Ridge/RLS)	High on ARM Cortex-M/FPGA

II. STRUCTURE OF THE COMPLETE REVIEW MANUSCRIPT

Section 1: Introduction & Academic Landscape | **Section 2:** BMS Estimation Tasks (SOC, SOH, SOP, RUL)
Section 3: Classical Model-Based Estimators & Multi-Timescale Decoupling | **Section 4:** Conventional Deep Learning Architectures
Section 5: Spiking Neural Networks for Battery Telemetry | **Section 6:** Reservoir Computing, Echo State Networks & LSMs
Section 7: Neuromorphic Silicon Benchmarking (Loihi 2, SpiNNaker2, Akida, Artix-7) | **Section 8:** Event-Driven Asynchronous Sensing Schemes
Section 9: Multi-Dimensional Benchmark & Radar Analysis | **Section 10:** Open Challenges & 5-Year Horizon | **Section 11:** Conclusion & Synthesis

*Note: The authoritative manuscript containing all mathematical formulations, 11 detailed sections, and complete reference bibliography is maintained in markdown format at **reference/review_paper.md**. This document serves as the formal IEEE preprint deliverable.*